RW–952

Sub. Code
536101

M.Sc. DEGREE EXAMINATION, NOVEMBER 2017

First Semester

Chemistry

CONCEPTUAL INORGANIC CHEMISTRY

(CBCS – 2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

1. Define bond order.

2. List out the different types of chemical bonds.

3. What is a buffer solution?

4. Give two examples of Lewis acids and bases.

5. What is FCC? Give an example.

6. What is an antifluorite structure?

7. Define Q-value.

8. What is Steller energy?

9. What are the stable oxidation states of actinides?

10. Write the position of lanthanides in the periodic table.

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

All questions carry equal marks.

11. (a) Explain

    (i) Dipole moment

    (ii) Born Haber cycle. (2.5 + 2.5)

    Or

    (b) Write a note on the following :

    (i) Electronegativity

    (ii) Lattice energy. (2 + 3)

12. (a) Write a brief note on acid strength of oxy acids.

    Or

    (b) Explain HSAB principle.

13. (a) Describe the structure of Zinc blends.

    Or

    (b) Explain the limiting radius ratio of tetrahedral, octahedral and cubic site.

14. (a) Explain the liquid drop model.

    Or

    (b) Discuss the utility of radioisotopes in neutron activation.

15. (a) What are the uses of lanthanides and actinides?

    Or

    (b) What are the spectral and magnetic properties of lanthanides?

Part C

$(3 \times 10 = 30)$

Answer any **three** questions.

All questions carry equal marks.

16. Explain the VSEPR theory in details.

17. What are the factors that influence the strength of acids and bases?

18. Explain the AB_2 types of crystals with suitable example.

19. Discuss α, β and γ modes of decay with examples.

20. Describe any three separation technique of lanthanides.

RW–953

Sub. Code
536102

M.Sc. DEGREE EXAMINATION, NOVEMBER 2017

First Semester

Chemistry

CONCEPTUAL ORGANIC CHEMISTRY

(CBCS – 2015 onwards)

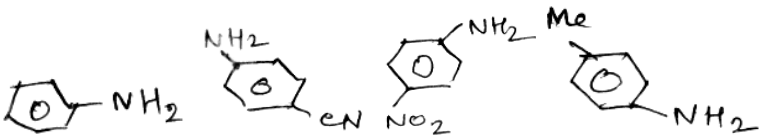
Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is hyperconjugation? What types of orbitals are involved in overlap in this phenomenon?

2. Compare the basicity of the following compounds :



3. Strength of the base employed can decide the elimination/substitution ratio. Illustrate with an example.

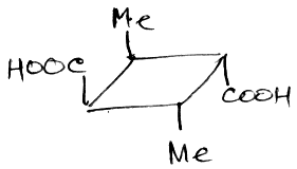
4. In what way, S_E2 and S_Ei mechanisms can be distinguished?

5. Indicate whether CF_3 group is activating or deactivating towards aromatic electrophilic substitution reaction. Is it ortho/para orienting or meta orienting?

6. What happens when sulphanilic acid is treated with excess bromine water? Suggest the mechanism of the reaction.

7. Predict the total number of possible stereoisomers for Ph CHBr CHBr CHBr Ph. Depict them all.

8. List out the symmetry elements present in the following compound :



9. What is Curtin-Hammett principle?

10. Distinguish between enantiotopic hydrogens and diastereotopic hydrogens.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) What is Huckel rule? Comment on aromaticity of annulenes taking [12]-annulenes and [18]-annulenes as examples.

Or

(b) Explain how benzyne can be trapped. Indicate any two reactions that go via benzyne intermediate.

12. (a) Write a note on neighbouring group participation.

Or

(b) Explain how solvent polarity and substrate structure influence the course of aliphatic nucleophilic substitution reactions.

13. (a) What is substituent constant? Describe the significance of the sign of the σ constant. Explain what is meant by σ_p and σ_m for one substituent.

Or

(b) What is Meisenheimer complex? Describe aromatic nucleophilic substitution reactions that involve Meisenheimer complex as intermediate. Discuss the mechanism in detail.

14. (a) Explain Cahn-Ingold-Prelog rules and indicate how they can be used to assign the R/S configuration of Chiral centres.

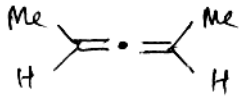
Or

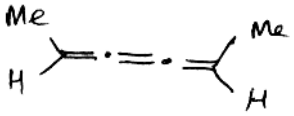
(b) What are diastereomers? How do they differ from enantiomers? Illustrate with examples. Indicate how geometrical isomers can also be considered as diastereomers.

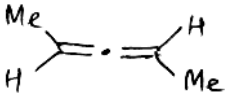
15. (a) Show that properly substituted biphenyls can exhibit enantiomerism. Discuss the origin of chirality in these cases.

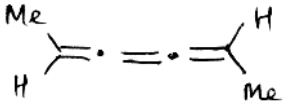
Or

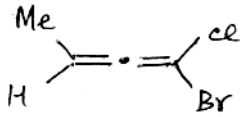
(b) Which of the following compounds are optically active?

(i) 

(ii) 

(iii) 

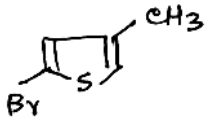
(iv) 

(v) 

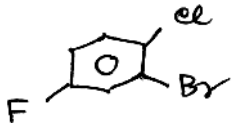
Part C (3 × 10 = 30)

Answer any **three** questions.

16. (a) Write the IUPAC name for the following compounds :

(i) 

(ii) CH_3 CHOH CHOH CH_2 CH_2 CHO

(iii) 

(b) Write a note on kinetic isotope effect.

(c) Illustrate Hammond postulate. (3 + 4 + 3)

17. Account for the following observations :

(a) Allylic halides undergo S_N1 reactions faster than ethyl halides

(b) Thiophenoxide anion is more powerful as a nucleophile compared to phenoxide

(c) More branched olefins are preferentially formed when alkyl halides undergo HX elimination

(d) E_2 reaction is stereospecific

(e) Torylate anion is a good leaving group. (5 × 2)

18. Describe the mechanism of

(a) Vilsmier Haack reaction

(b) Echichibabin reaction and

(c) Gattermann Koch reaction. (4 + 3 + 3)

19. Write notes on :

(a) E, Z-nomenclature

(b) Differences between chiral centre, chiral plane and chiral axis

(c) Resolution of racemic mixture. (3 + 4 + 3)

20. (a) Compare the stability of chair form of cyclohexane and boat form of cyclohexane.

(b) Explain why trans decalin is rigid, while cis-decalin is flexible.

(c) Discuss the stereochemical outcome of E2 elimination of HBr from meso-2, 3-dibromobutane.

(d) What are ansa compounds? (3 + 3 + 2 + 2)

RW–954

Sub. Code
536103

M.Sc. DEGREE EXAMINATION, NOVEMBER 2017

First Semester

Chemistry

CONCEPTUAL PHYSICAL CHEMISTRY

(CBCS – 2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. State Heisenberg's uncertainty principle.

2. What is Hermitian operator?

3. What are polarizable electrodes?

4. Write the equation relating charge-density and interfacial tension.

5. Distinguish between rate and rate constant.

6. What is steady state approximation?

7. What are irreversible processes? Give an example.

8. Classify various phase transitions.

9. What is resonance energy transfer?

10. What are photovoltaic cells?

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Explain probability density of waves.

Or

(b) What are commutators?

12. (a) Explain Helmholtz-Perrin model of double layer.

Or

(b) Derive Tafel equation.

13. (a) How will you determine rate law?

Or

(b) Explain kinetic isotopic effect.

14. (a) Write the fundamental equations for open systems.

Or

(b) Write the applications of Debye-Huckel theory.

15. (a) Explain photosensitization with mechanism.

Or

(b) Explain the concept of photocatalysis.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss various quantum mechanical operators.

17. Derive Butler-Volmer equation.

18. Explain Lindemann-Hinshelwood mechanism.

19. Discuss the thermodynamics of ideal solutions.

20. How will you measure the efficiency of a solar cell?

RW–955

Sub. Code
536501

M.Sc. DEGREE EXAMINATION, NOVEMBER 2017

First Semester

Chemistry

Elective I — ANALYTICAL CHEMISTRY

(CBCS – 2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is meant by precision?

2. List out the sources of systematic errors.

3. What is the principle of chromatography?

4. How is column packed in GC?

5. What are the types of membranes?

6. What are the factors controlling lonic separation?

7. What are the principle of thermal analysis?

8. What is the limitation of flame photometry?

9. What are the differences between electrochemical sensors and biosensors?

10. What are the advantages of DME used in polarography?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain the following terms with proper examples.

    (i) Mean

    (ii) Median

    (iii) Average deviation

    (iv) Standard deviation.

    Or

    (b) Explain significant figures. What is the rule to determine significant figures?

12. (a) Explain ascending and descending methods of paper chromatography.

    Or

    (b) Write a short notes on TLC.

13. (a) Explain the Reverse Osmosis Process.

    Or

    (b) What are the types of equipments employed for electrophoresis, dielectrophoresis and electro dialysis separations?

14. (a) Compare AAS with AES.

    Or

    (b) Write a brief note on thermometric titrations.

15. (a) Write a short note on cyclic voltametry and stripping voltametry.

    Or

    (b) What are the different types of amperometric titration curves?

Part C

(3 × 10 = 30)

Answer any **three** questions.

16. What is meant by error? Provide the different types of errors and indicate how it can be minimized.

17. Explain briefly on :

    (a) Gas chromatography

    (b) HPLC.

18. Explain the following :

    (a) Nanofiltration

    (b) Donnan – dialysis

    (c) Choice of adsorbents for adsorption techniques

    (d) Microfiltration

    (e) Ultra – filteration.

19. Explain :

    (a) Spectrofluorimetry

    (b) Turbidimetry

    (c) Nephelometry.

20. Write notes on :

    (a) Polarographic maxima

    (b) Half-wave potential

    (c) Dissolved oxygen.

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RW-956

Sub. Code
536502

M.Sc. DEGREE EXAMINATION, NOVEMBER 2017

First Semester

Chemistry

ENVIRONMENTAL AND GREEN CHEMISTRY

(CBCS – 2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

All question carry equal marks

1. What is acid rain?

2. What is potable water?

3. What is break point chlorination?

4. What is electro dialysis?

5. Write a note on "designing of safer chemicals".

6. How will you select appropriate anxillians substances?

7. What is supercritical CO_2?

8. List two example for biocatalyst.

9. What is meant by carbon foot print?

10. Write a note on cerbon sequestration.

Part B (5 × 5 = 25)

Answer **all** questions. choosing either (a) or (b)

All questions carry equal marks

11. (a) What is green house effect? What are its consequences? What steps do you suggest to control it?

Or

(b) How will you determine BOD, micronutrients and heavy metals in water?

12. (a) Write a note on :

(i) Aeralied lagoons

(ii) Oxidation pand

Or

(b) Describe ultra filtration method what are the advantages and disadvantages of this method?

13. (a) State the twelve principles of green chemistry.

Or

(b) Discuss the development of analytical techniques to prevent the generation of hazardows substances.

14. (a) Discuss the synthesis of caprolactum by using principles of green chemistry.

Or

(b) Explain the coupling reactions using ultra sound.

15. (a) Discuss microwave assisted reactions in organic solvents.

Or

(b) Explain the microwave assisted reaction of oxidation of toluene to benzoic acid.

Part C (3 × 10 = 30)

Answer any **three** questions.

All questions carry equal marks

16. What are the sources of water pollution? Explain how will you determine SS, VOC, TDS in weter?

17. Write a note on :

    (a) Disinfection method

    (b) Sedimentation

    (c) Coagulation of water treatment.

18. Compare atom economy reactions and uneconomy reaction with suitable examples.

19. Explain ultrasound assisted reactions involving.

    (a) Estrification and

    (b) Reaction.

20. Discuss reaction an solid support and solvent free esters saponifications.

RW–957

Sub. Code
536301

M.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Third Semester

Chemistry

CHEMISTRY OF INORGANIC COMPLEXES

(CBCS – 2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. What are the factors affecting CFSE?

2. What are the weak field and strong field ligand? Give one example for each.

3. Explain the annation reaction.

4. Discuss the π -bonding theory.

5. Write the ground state term symbol for V^{3+} complexes.

6. Explain the electronic spectra of Mn^{2+} complexes.

7. What are the medical application of MRI-agents?

8. Draw the structure of any one of the porphyrin ring system.

9. What is the Biological function of myoglobin?

10. What are rubredoxins? What are its biological significance?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Factors affecting the magnitude of 10D9?

Or

(b) Bring out the comparison between CFT and MOT.

12. (a) Explain the substitution reaction in square planar complex.

Or

(b) What are the factors affecting the acid hydrolysis?

13. (a) What are the differences between orgel and tanabe sugano diagrams?

Or

(b) Discuss the magnetic properties of co-ordination compounds.

14. (a) Discuss the potential metal binding sites of any two bio-ligands.

Or

(b) Discuss the toxcity of metals, Hg and Cr with suitable example.

15. (a) Explain the structure and functions of cytochrome.

Or

(b) Discuss the structure and properties of hemocynin.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. What is CFT? Write the merits and demerits of crystal field theory complexes.

17. Explain the mechanism of outersphere redox reaction of co-ordination compounds.

18. Write a note on the experimental determination of magnetic susceptibility by using Gouy balance.

19. Sketch and explain the mechanism of sodium pumping.

20. Discuss the structure and biological role of Hemoglobin.

RW–958

Sub. Code
536302

M.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Third Semester

Chemistry

MODERN METHODOLOGIES IN ORGANIC CHEMISTRY

(CBCS – 2015 onwards)

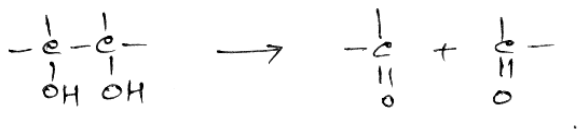
Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

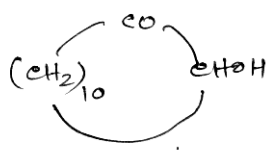
Answer **all** questions.

1. Explain how olefins can be converted to epoxides?

2. Provide the list of reagents that can effect the following conversion :



3. Suggest a method of preparing the following compound :



4. Illustrate the enantioselective reduction with an example.

5. What is Henry reaction?

6. Explain the role of phase transfer catalysts.

7. Indicate how 1,3 – diols can be protected.

8. Explain how phthalic anhydride can act as a good protecting group for primary amines?

9. Perform one bond disconnection on longifolene in all possible ways and draw the pretargets in each case.

10. By logic, compare the enantiomeric excess that could results from the third generation asymmetric synthesis and fourth generation asymmetric synthesis.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

All questions carry equal marks.

11. (a) Write a note on Woodward and prevost hydroxylation process.

Or

(b) What is hypervalent iodine? Explain the range of hypervalent iodine reagents available for different oxidation processes.

12. (a) Describe how lithium aluminum hydride acts as a versatile reagent effecting powerful reduction of different functional groups.

Or

(b) Discuss the role of noble metals in heterogenous catalysis. Describe the mechanism of catalytic reductions.

13. (a) Describe the principle, mechanism and applications of Suzuki reaction.

Or

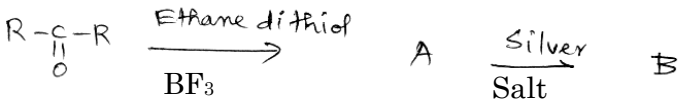
(b) Explain the synthetic utilities of Baker's yeast and Merrifield resin.

14. (a) (i) How can a e=e can be protected during synthetic transformation?

(ii) How can $ArNH_2$ be converted to ArCN and ArF?

Or

(b) (i) Predict the course of the following conversion :



(ii) How can the following conversion be effected?

15. (a) Describe the synthesis of jasmone.

Or

(b) How can the side chain of retinol be built up?

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

All questions carry equal marks.

16. Write notes on :

(a) Sharpless asymmetric epoxidation

(b) Bayer Villiger oxidation.

17. (a) Explain the mechanism of formation of cyclohexadiene by Birch reduction.

(b) Explain the selectivity associated with MPV reduction.

(c) Illustrate Nef reaction.

18. Explain the need for retrosynthetic analysis. Discuss the terms involved in retrosynthesis. Discuss the strategies to be followed during retrosynthetic approach.

19. Write notes on :

(a) Robinson annulation

(b) Nazarov reaction

(c) Baylis Hilman reaction.

20. Describe the steps involved in the synthesis of cholesterol.

RW–959

Sub. Code
536303

M.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Third Semester

Chemistry

ADVANCED PHYSICAL CHEMISTRY

(CBCS – 2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. Distinguish between ionization energy and electron affinity.

2. What are quantum defects?

3. What are eigen values and eigen states?

4. What are vertical transitions?

5. Explain various components of a battery.

6. What is the principle of fuel cell?

7. Write Bose – Einstein equation and explain the terms.

8. Write the translational partition function.

9. Write Bragg's law and explain the terms.

10. What are intrinsic semiconductors?

Part B $(5 \times 5 = 25)$

Answer **all** the questions, choosing either (a) or (b).

11. (a) How will you separate internal motion and radial solution?

Or

(b) Write the importance of term symbols.

12. (a) Discuss the role of rotational level degeneracy.

Or

(b) Discuss the polarizability and selection rules for rotational and vibrational Roman spectra.

13. (a) Describe the function of Leclanche cell.

Or

(b) Explain Pourbaix diagram.

14. (a) Write notes on grand canonical ensembles.

Or

(b) Discuss linear response theory.

15. (a) Explain Frankel defect.

Or

(b) Write Kroger – Vink notation for crystal defects.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss the various atomic orbitals and their energies.

17. Discuss the theory and principles of microwave spectroscopy.

18. Discuss the theory and principles of fuel cells.

19. Derive Bose – Einstein statistics.

20. Discuss the band theory of solids.

RW–960

Sub. Code
536304

M.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Third Semester

Chemistry

SPECTROSCOPY – APPLICATIONS OF ORGANIC AND INORGANIC CHEMISTRY

(CBCS – 2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Explain the terms overtone and combination band.

2. Describe how the primary and secondary amines can be distinguished by IR spectroscopy.

3. Explain $\gamma -$ gauche effect.

4. Sketch the off-resonance decoupled ^{13}C NMR spectrum of $CH_3CH_2COCH(CH_3)_2$.

5. Explain the origin of four lines in the ^{19}F NMR spectrum of methye fluoride.

6. What is hyperfine splitting constant? Explain.

7. What is soft ionisation technique?

8. Illustrate McLafferty rearrangement.

9. For a hydrocarbon, the intensity of $(M+1)^+$ ion peak is 6.6% of that of the molecular ion peak. How many carbons are there in the compound?

10. The organic molecule C_3H_8O has no peak in the region of $2900\ cm^{-1}$ to $4000\ cm^{-1}$. What is the structural formula of this compound?

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

All questions carry equal marks.

11. (a) Discuss the effect of conjugation, ring size and hydrogen bonding on the stretching frequency of the carbonyl group.

Or

(b) Describe the principle and application of Raman Spectroscopy. Indicate how it is complimentary to IR spectroscopy.

12. (a) Explain the applications of COSY and DEPT spectra.

Or

(b) Mention the role of (i) shift reagents and (ii) decoupling in simplifying complex NMR spectra.

13. (a) Describe how Dynamic NMR spectroscopy can help to arrive at the thermodynamic parameters of the equilibrium and exchange processes.

Or

(b) ESR spectroscopy has wide applications in the structure analysis of inorganic complexes. Illustrate.

14. (a) Describe the principle and theory of Mössbauer spectroscopy.

Or

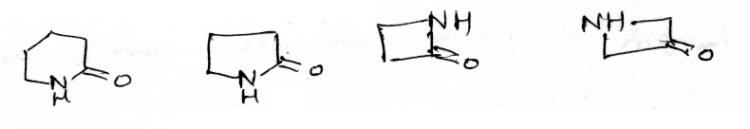
(b) What is the difference between molecular ion peak and base peak? What are the factors that decide the intensity of the molecular ion peak in EI-MS spectra?

15. (a) The compound with molecular formula $C_8H_{10}O$ has a UV band at 20 nm. The IR spectrum has bands at 3350, 3000, 2985, 1610, 1580 and 1450 cm^{-1}. The ^{13}C off-resonance NMR has one quartet, four doublets and one singlet. The PMR spectrum has the following features :

130 $\delta(3H, d)$ 4.91 $\delta(1H, q)$; what is the structure of this compound?

Or

(b) Arrange the following compounds in the increasing order of their carbonyl stretching frequency :



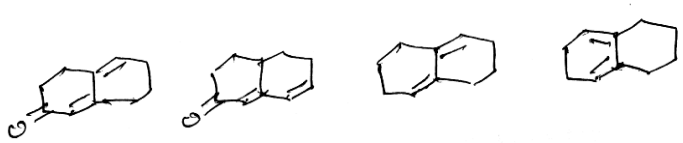
Predict the 'H NMR pattern for each.

Part C (3 × 10 = 30)

Answer any **three** questions.

All questions carry equal marks.

16. Discuss the Woodword Fieser rules as applicable to enones. Explain how the solvent polarity influences the $n \rightarrow \pi^*$ as well as $\pi \rightarrow \pi *$ absorption positions. Calculate the λ max for the following compounds :



17. (a) Discuss the factors that affect the chemical shift of hydrogen and e-13 nuclei.

(b) Explain the factors that affect the magnitude of vicinal coupling constants.

18. Write notes on : (a) Zeemna effect (b) NMR spectra of boron compounds (c) Magnetic susceptibility measurements.

19. Write notes on : (a) MALDI-MS (b) Isotope peaks in mass spectrometry (c) Mass fragmentation of aliphatic and aromatic nitro compounds.

20. Four isomeric compounds (A, B, C and D) with molecular formula C_6H_{12} have the following proton NMR features. Find out the compounds.

Compound A : Only one signal at 1.40 ppm.

Compound B : Only one signal at 1.66 ppm.

Compound C : 1.0 $\delta(s, 9H)$; 5.8 $(1H, d)$; 5.9 $(1H, d)$
4.8 δ $(1H, dd)$.

Compound D : 1.2 $\delta(6H, d)$; 1.8 $(3H, s)$; 2.2 $(1H,$ septet$)$
4.6 $\delta(1H, d)$; 4.7 $(1H, d)$.

Compound A has two carbon-13 signals, while compound B has only one carbon-13 signal.

RW–961

Sub. Code
536703

M.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Third Semester

Chemistry

ANALYTICAL INSTRUMENTATION

(CBCS – 2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions in about 100 words each.

1. State Bragg's law.

2. Write any two advantages of X-ray method.

3. Give the basic principle of NMR.

4. What is hyper Raman effect?

5. What is the essential requirement for a solvent to be used in UV Spectroscopy?

6. Why is ethanol a good solvent in UV?

7. Distinguish between TGA and DSC.

8. Define change in enthalpy.

9. What is the principle of AFM?

10. What is the principle of SEM?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) List out the limitations of X-ray diffraction.

Or

(b) Write a short note on X-ray detectors.

12. (a) Describe the working principle of Laser Raman Spectrometer.

Or

(b) With neat diagram explain the working principle of NMR and sample preparation techniques.

13. (a) Explain the applications of UV absorption spectrometry.

Or

(b) Explain in detail applications of photoluminescence spectrometry.

14. (a) With neat block diagram explain working principle of Differential Scanning Calorimetry.

Or

(b) Discuss in detail the mechanical principles of DSC.

15. (a) Explain the working principle of AFM and calculation of RMS value.

Or

(b) Write a short note on contact and non-contact mode of AFM.

Part C

(3 × 10 = 30)

Answer any **three** questions.

16. Explain the principle and working of X-ray fluorescence spectrometer.

17. With neat illustration explain the working principle of ESR spectrometer.

18. What are filters? Enumerate the role of filters and detectors in UV-absorption spectrometry.

19. Explain the working principle of TGA.

20. Describe the principle, construction and working of ESCA.